

Multi-year analysis of electrified storm cells over north Alabama

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ABSTRACT

Multi-year analysis is often used to understand how characteristics generally relate to each other. In meteorology, general characteristic studies can, for example, be applied to how supercell size relates to tornado intensity or how stratiform reflectivity relates to rainfall rates. This study was designed to see if it would be possible to observe lightning and storm cell characteristics over a large time period for a more thorough understanding of the relationship. Analyzing lightning characteristics within a multi-day event has been done for severe events across the world, but the ability to analyze years of data can provide a general idea of how storms impact lightning over a specific region. This project made use of Adapt, an orchestrated framework for total analysis of a storm cell, alongside xlma-python, a program for viewing lightning data from a lightning mapping array network. After analyzing 153 events from 2019-2026 over north Alabama, over 65000 storm tracks with 1.2 million flashes were identified. Patterns in the data were observed relating increases in lightning size and strength to differential reflectivity values between 0-2 db but with opposite reflectivity values. An increasing trend was also identified where lightning activity increased following storm identification, even when the storms themselves do not appear to strengthen. The results of this study agree with other results regarding which storm processes modify lightning characteristics, while proving that this kind of analysis is possible with large datasets and can be performed across multiple regions. The large dataset allowed a unique case to be identified with little lightning when radar characteristics would indicate the opposite. Further developing Adapt into a thorough analysis tool allowing for association of measurements by different instruments with storm cells will continue to make larger analysis worthwhile and improve forecasting and early warning techniques through larger studies.

26 **1. Introduction**

27 Lightning is known to be the result of large positive and negative charge buildup within severe
28 storms with strong updrafts. These strong updrafts cause increased collisions between ice crystals
29 and graupel, which separates charges into regions based on the hydrometeor's density (Cooray
30 2014). When charge concentrations reach significant levels, lightning is initiated and the charge
31 is dispersed. Knowing the factors that influence particle collisions is helpful in forecasting when
32 and where lightning will occur. Environments with a lot of instability tend to lend themselves to
33 stronger updrafts, but there does need to be adequate moisture as well. Unfortunately, the explosive
34 nature of severe storms tends to be difficult to observe because of how much can change between
35 radar scans, but lightning detection systems will often report data before a radar does and the rate
36 at which flashes occur can tell a forecaster if the storm cell is strengthen or weaken before the
37 radar does. Work reported in Kingfield et al. (2017) and others has shown that not only are tall
38 structures more susceptible to lightning strikes, but the tall structures will often attract lightning
39 that would have struck out to 1 km out, depending on the height of the tower. This is even more
40 true for rotating objects, such as wind turbines, which can attract lightning from further away if
41 they are spinning. When it comes to wind farms, knowing when lightning will strike can be the
42 difference between a busted and a safe turbine.

43 The correlation between different polarimetric radar variables and the occurrence of lighting has
44 had many studies performed on a case study basis for different environments, but studying both
45 over multiple years has seldom been achieved. Multi-year CONUS studies of radar data include
46 Williams et al. (2022) and Farby et al. (2017) where both authors used radar mosaics from the
47 WSR-88D radar network to identify rain fall climatologies or other types of severe weather. The
48 data ranged from 1998-2011 and 1996-2015, respectively. Fuchs et al. (2016) is one of the few

49 multi-year lightning studies that have been performed, within which the Washington D. C., north
50 Alabama, and northeast Colorado lightning mapping array (LMA) networks were used. The LMA
51 time periods ranged between 2007-2014 and mapped over 63 million flashes. Fuchs et al. also
52 made use of the National Lightning Detection Network (NLDN) and satellite-based observations
53 to compare the accuracy of all three networks.

54 This study aimed to take use the results of both types of datasets to perform a full-scale analysis
55 of electrified storm cells over one single area, which could later be expanded to more regions. We
56 analyzed data from June 1st, 2019 to February 27th, 2026 over north Alabama and found over 1
57 million flashes across more then 13000 electrified storm cells. Specific radar characteristics that
58 are known to have a relation to lightning characteristics were plotted to see if the dataset agreed
59 with prior results. Analysis of a single event that did not match expected characteristics was also
60 performed.

61 **2. Data and Methods**

62 This study focused on North Alabama due to the proximity of a Lightning Mapping Array
63 (LMA) network and a WSR-88D Radar (KHTX) in Huntsville, AL to an Atmospheric Radiation
64 Measurement (ARM) site in Bankhead National Forest (BNF), about 80 km to the SW. The LMA
65 network includes 11 stations centered on Huntsville while KHTX is located about 50 km to the
66 ENE. Table 1 shows the time domain of the study and initial statistics of the data analysed. All
67 of the radar analysis was made possible by ARM-Adapt which tracked storm cells and processed
68 characteristics across multiple radar scans, creating a database for the full lifecycle of each cell. The
69 lightning analysis was done through xlma-python which took the unfiltered LMA data, processed
70 and removed any erroneous data, and clustered each recorded source into flashes. The filtering
71 criteria used in both can be seen in Table 2.

72 While there are studies looking at lightning characteristics on a multi-year scale and numerous
73 studies for storm cell characteristics at the same level, there have been very few studies for both
74 (Fuchs et al. 2016). The projects that have looked at the relationships between storm cell and
75 lightning characteristics have been done for one season or less (Brown et al. 2002, Lyons and Keen
76 1994, Utsav et al. 2022). While the goal was to have similar sample sizes for each part of the
77 year, May through September included substantially more storm cells which likely skewed the data
78 towards a warm-season bias (Fig. 1, Table 3).

79 **3. Results and Discussion**

80 Table 1 provides an immediate overview of the data analyzed but intentionally neglects to mention
81 some key distinctions. Of the over 65000 cells analyzed, 13754 cells (21%) were found to have
82 lightning occurring. These near-14000 cells were used for this study. Figure 2a visualizes an
83 average of cell echo top and bottom heights at 10 dBz intervals, for every cell recorded. The
84 beginning of a cell is determined by when it first reached the Adapt thresholds set in Table 2,
85 either by initiating and growing within the domain or propagating into the domain having initiated
86 further away. The end of a cell is determined when the cell either goes below the threshold for
87 multiple successive scans or propagates outside of the domain. The black line is a visualization of
88 the number of flashes happening at any minute within a cell's lifespan. Figure 2b shows the total
89 number of cells for each time after identification on a logarithmic scale, starting with over 10000
90 cells and falling below 50 around 9000 seconds after identification.

91 Figure 2 provides an understanding of how a cell's life cycle impacts the amount of lightning
92 occurring. While at the initial scan, there are only a handful of flashes occurring within a cell, that
93 number quickly jumps to around 7 flashes for most of the cell's life, before decreasing after 8000
94 seconds. The variation in echo height regions is also relatively stable where most of the initial

95 identification scans show the higher reflectivity regions to be higher AGL, there again is not much
96 variation until around 8000 seconds where every region starts to more quickly decrease in height.
97 The 10 and 50 dBz regions appear to have the most drastic changes, and the flash rates reflect that,
98 decreasing at a similar time. While these phenomena could be related to a sample size issue since
99 there are less than 100 cells by 8000 seconds, even just the 50 cells that occurred during multiple
100 events should still provide a good understanding of the storm structure that far into a storm's life
101 cycle.

102 Figure 3 and Figure 4 show similar patterns but each provides a unique understanding of how
103 differential reflectivity relates to lightning. Both showcase the differential reflectivity mean for
104 a cell alongside the maximum flash area and maximum flash rate, respectively. Both also have
105 a peak between 1 and 2 dB, but the maximum reflectivity that colors each point shows that the
106 combination of differential reflectivity and reflectivity indicate different lightning characteristics.
107 Figure I (appendix) shows how flash area and flash rate relate to each other . . . Figure 3 highlights
108 the lower reflectivity and differential reflectivity relationship, which suggests those cells were
109 composed of primarily of many densely packed, medium to large sized, rain drops indicative of a
110 stratiform precipitation region. Works such as Peterson et al. 2022 have found larger flashes happen
111 more often within stratiform precipitation regions due to the large charge regions with slower charge
112 accumulation. Conversely, Fig. 4 is showing the relationship between mean differential reflectivity
113 and flash rate per minute. The results also support the literature, with differential reflectivity
114 between 1-2 and reflectivity in the 50-60 dBz range, indicative of larger raindrops and wet hail.
115 The storms conducive to hail and large drops have stronger updraft speeds than cells without hail
116 formation and a stronger updraft causes more collision charging to occur which causes stronger
117 charge regions which causes higher flash rates.

Both of these results validate the current knowledge of how the structure of a cell corresponds to different lightning characteristics and the multi-year overview assists to verify the case studies that have been performed in the literature. There is one oddity in these plots that requires its own study, but the cluster of reflectivity maximum over 65 dBz, low differential reflectivity, and low flash characteristics all occur during one event that had less lightning than anticipated. The event occurred at the end of February, 2026 but the surface temperatures were around 20°C so this was undoubtedly a rain event. Figure 5 shows the radar scan where Adapt identified 29 cells at once at 02:01:36 Z in what would be expected to be a high-flash rate environment. Instead, the Adapt output shows less than 10 cells managed to achieve a flash count higher than 50 for one scan interval. In comparison, the average number of flashes produced by cells with reflectivity between 60-61 dBz was 130 flashes, for 46 cells.

4. Conclusions

This study sought to gain an understanding of how storm cells characteristics generally relate to lightning characteristics through a multi-year analysis of data recorded by the Huntsville, AL WSR-88D radar and the North Alabama Lightning Mapping Array. Adapt managed to process 153 events including over 65000 storm cells and 1.2 million flashes across a seven-year period. Over 13500 of the storm cells included lightning activity and those were what was focused on in this paper.

Figure 2a showed how the dataset can be used to display the average characteristics of every storm cell at once and the relation to lightning characteristics. Most notable, the data showed how storms and the number of flashes tend to grow until about 8000 s after identification, where most cells will then begin to shrink. For this project, an overall trend works to display what is possible,

140 but future studies may seek to narrow down the criteria for what cells are studied to only get data
141 on lightning rates in, for example, stratiform regions or tornadic supercells.

142 Figure 3 and Figure 4 show the possibility of individual characteristic analysis between previously
143 known connections. The relationship between reflectivity and differential reflectivity can highlight
144 small differences in the structure of a storm. Low reflectivity with low differential reflectivity can
145 mean that there are more drops but they appear smaller which can indicate a weaker updraft while
146 high reflectivity and high differential reflectivity will typically indicate very heavy rain or heavy
147 rain mixed with wet, melting hail. The combination of reflectivity and differential reflectivity was
148 used alongside flash rate and flash area to validate what previous studies have found in higher
149 flash rates with stronger updrafts and larger flashes in lower reflectivity, dense drop environments.
150 Future studies can build upon what was previously mentioned and filter the dataset to only include
151 specific storm types and get a better idea of what process within which storms assist specific
152 lightning characteristics.

153 The event first alluded to in Figures 3 and 4, but highlighted in Figure 5, was a clear anomaly in
154 the dataset, but still an important case when studying how storm characteristics relate to lightning.
155 While the rest of the dataset agreed with prior studies, this outlier was easy to identify because of
156 how Adapt allowed for more accessible analysis of storm cells across numerous events. This event
157 is an excellent example of identifying characteristics that could be narrowed in on to find other
158 events with similar storm regimes and gain a better understanding of the processes leading to the
159 unique properties.

160 This study highlighted two pieces of the larger Adapt dataset, a wholistic view of numerous cell's
161 life cycle and the unique variables that can be pulled out to reveal correlations that have been seldom
162 observed across multiple events. With Adapt recording min, max, and mean of every radar variable
163 alongside volume, speed, direction, and lifespan and the incorporation of variables from xlma-

python to get lightning flash counts, rates, sizes, and energy, analyzing storms of any combination of characteristics has become easier to achieve over large datasets. With the addition of more post process modules planned, the storm analysis capabilities of Adapt are seemingly endless. Future studies may seek to analyze specific storm regimes across a region or study multiple regions to see if generalizations can be made about all storms, instead of just storms in one climate.

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232 Peterson et al. (2022)

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239		study. The xlma-python criteria were based on sensitivity testing and the
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242 TABLE 1. Summary statistics for the study period. Start and End Time are given in UTC. Totals indicate the
243 number of events, storm cells, and lightning flashes included in the analysis.

Start Time	End Time	Total Time	Total Events	Total Cells	Total Flashes
5 Jun 2019 12:32 UTC	27 Feb 2026 04:40 UTC	6 years, 269 days, 16 hours, 8 minutes	153	65,696	1,183,663

TABLE 2. Filtering criteria for ADAPT storm cells and xlma-python flashes. ADAPT criteria were determined through sensitivity testing conducted prior to this study. The xlma-python criteria were based on sensitivity testing and the values recommended by MacGorman et al. (2008).

Software	Parameter 1	Parameter 2	Parameter 3	Parameter 4	Parameter 5
ADAPT	Minimum Cell Area	Maximum Cell Area	dBZ Threshold	Horizontal Distance Maximum	Vertical Distance Maximum
	5 km ²	800 km ²	40 dBZ	150 km	40 km
xlma-python	Minimum χ^2	Minimum Number of Stations	Minimum Number of Sources	Time Between Flashes	Distance Between Flashes
	1	6	5	0.15 s	3 km

TABLE 3. Number of storm events analysed per month.

Month	Number of Events
January	4
February	3
March	8
April	6
May	16
June	22
July	34
August	29
September	21
October	2
November	2
December	4

247

LIST OF FIGURES

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Fig. 1. Bar chart showing the number of storm cells present in each month of the year. The orange bars represent cells that did not have any lightning associated while the blue bars represent cells with lightning associated. 18

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Fig. 2. An average representation of the echo heights for every 10 dBz interval by each radar scan the cell was alive. The colors represent each echo height region and are defined in the key while the black line represents the average number of lightning flashes occurring at every minute after identification until the cell dissipates. 19

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Fig. 3. The total number of cells that exist for consecutive radar scans is shown on a logarithmic scale. Only cells with more than three scans and less than 70 are used for these two figures to limit the amount of noise. 20

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Fig. 4. A scatter plot of max differential reflectivity vs max flash area of each individual cell. Dots are colored by the cell's respective max reflectivity. 21

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Fig. 5. Same as Fig. 3, but for max flash rate instead of flash area. 22

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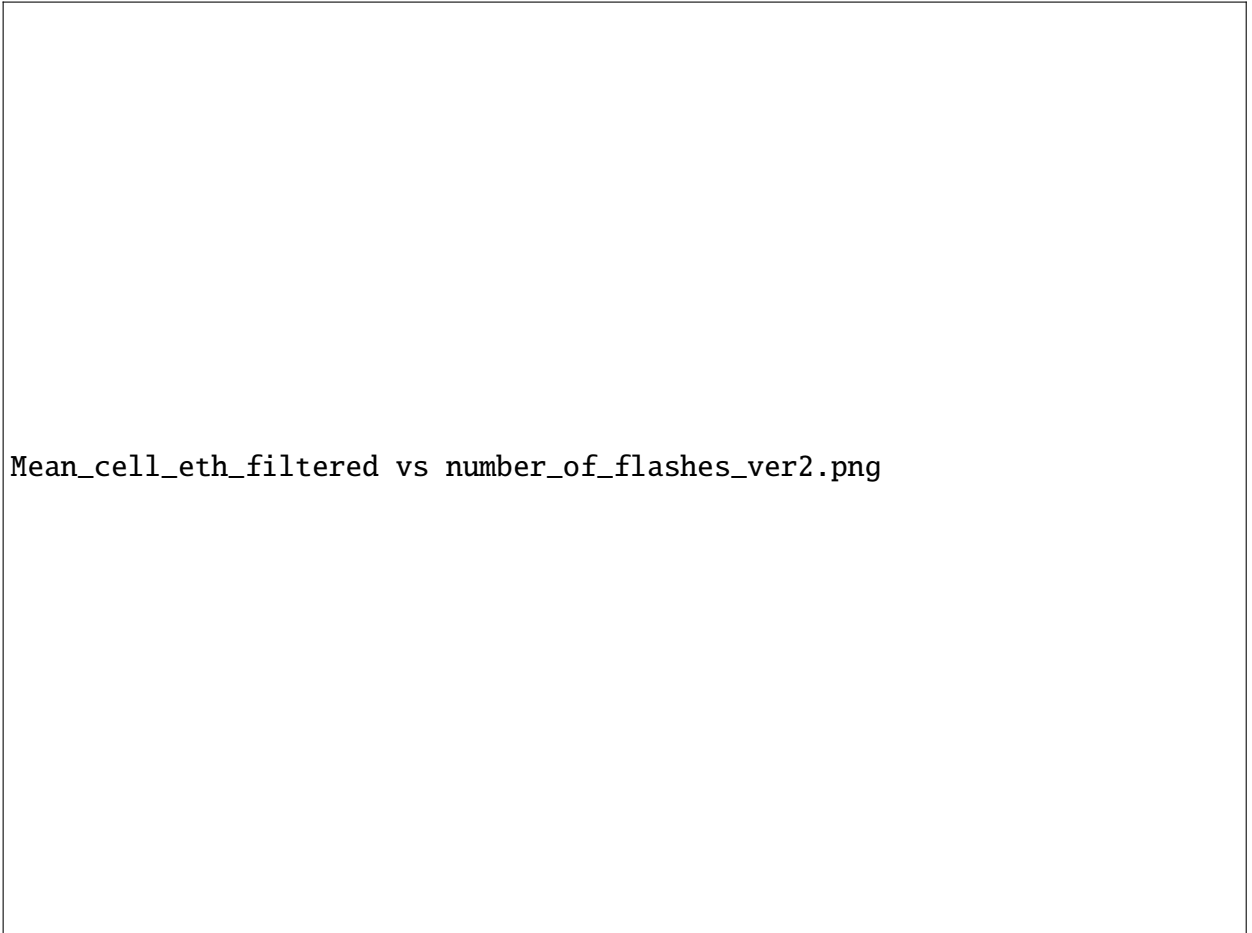
Fig. 6. Adapt output plot showing the reflectivity of the KHTX region on the left including derived motion vectors. On the right is the Adapt cell segmentation and projections for future scans within the minimum 40 dBz threshold. 23

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Fig. 7. Enter Caption 24



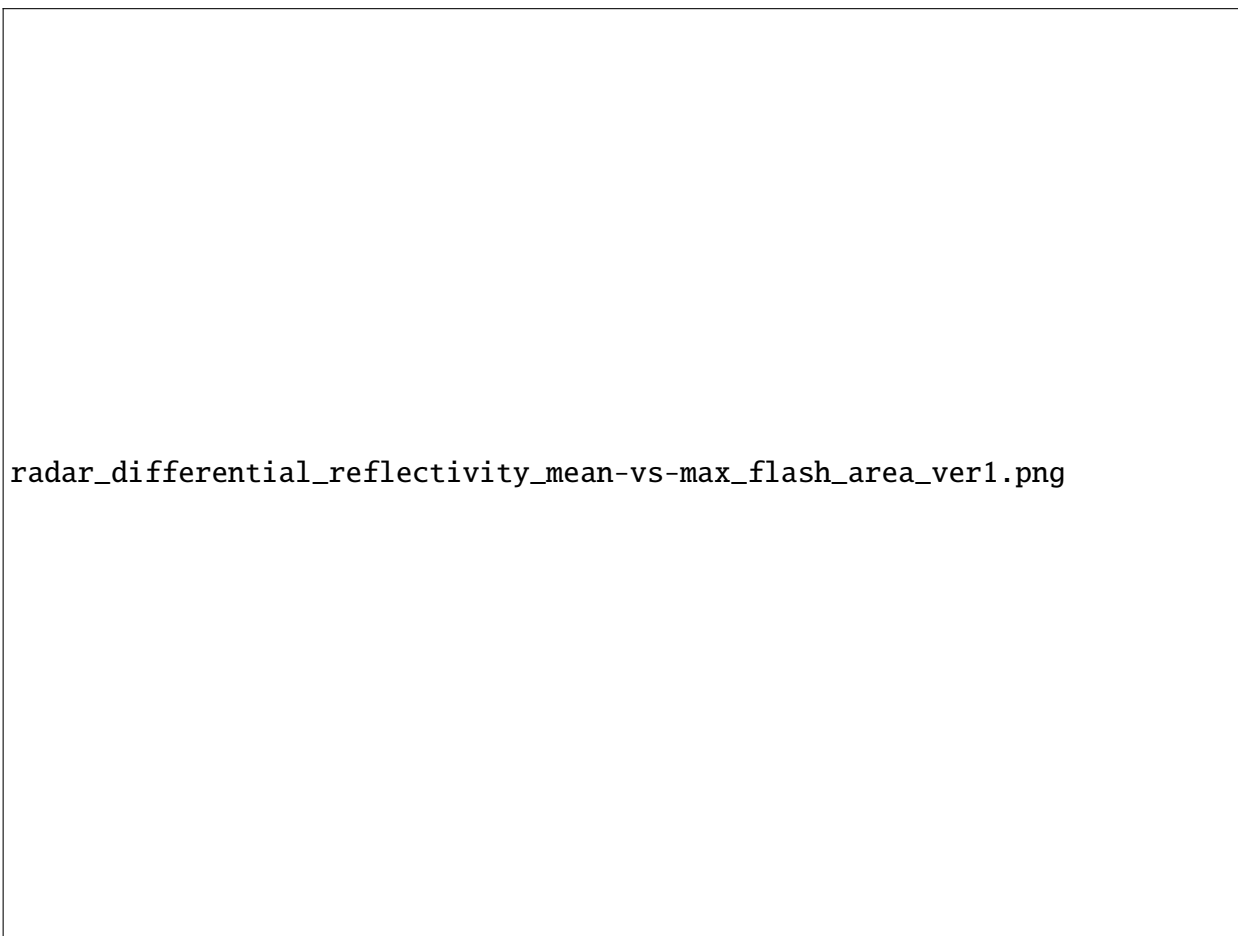
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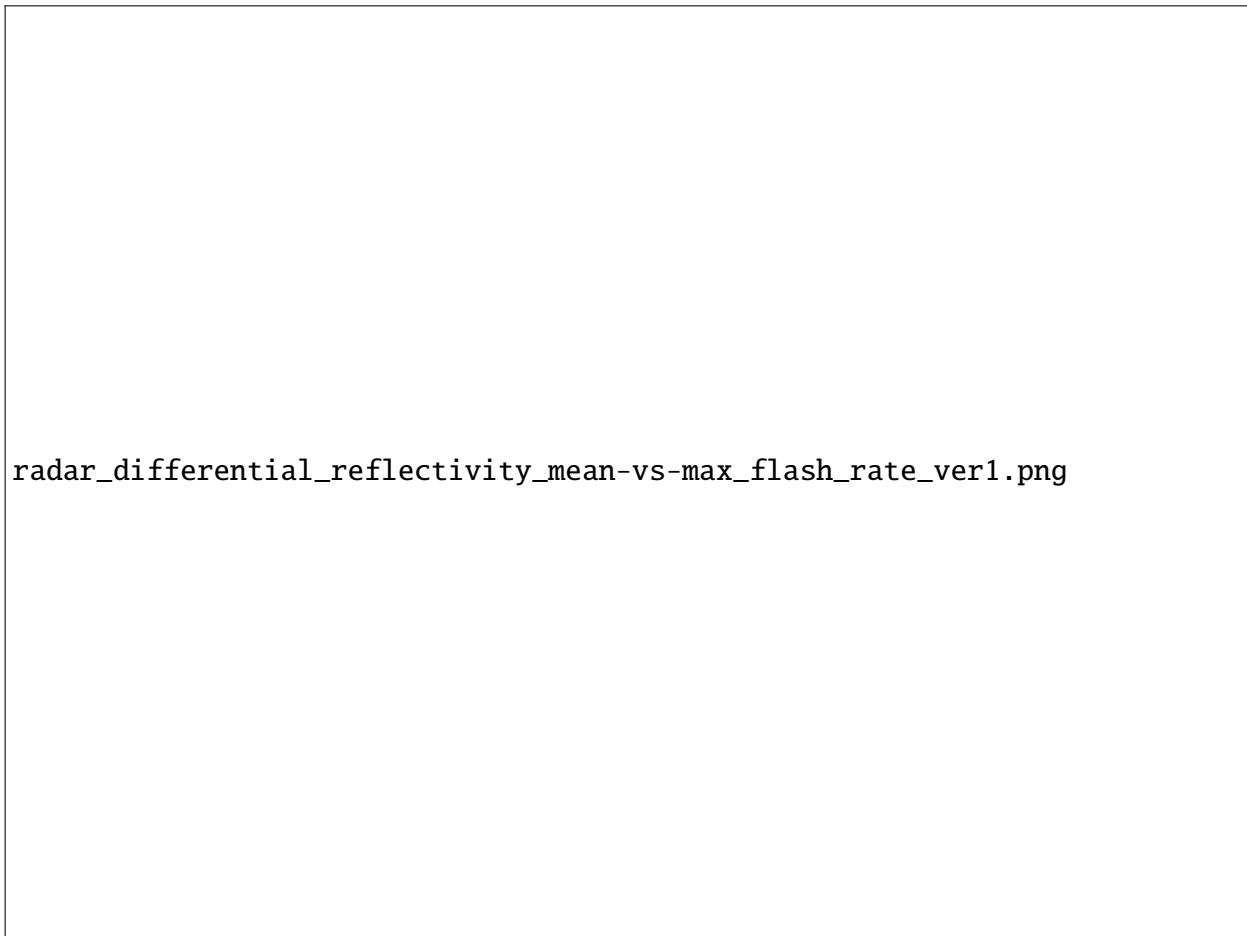
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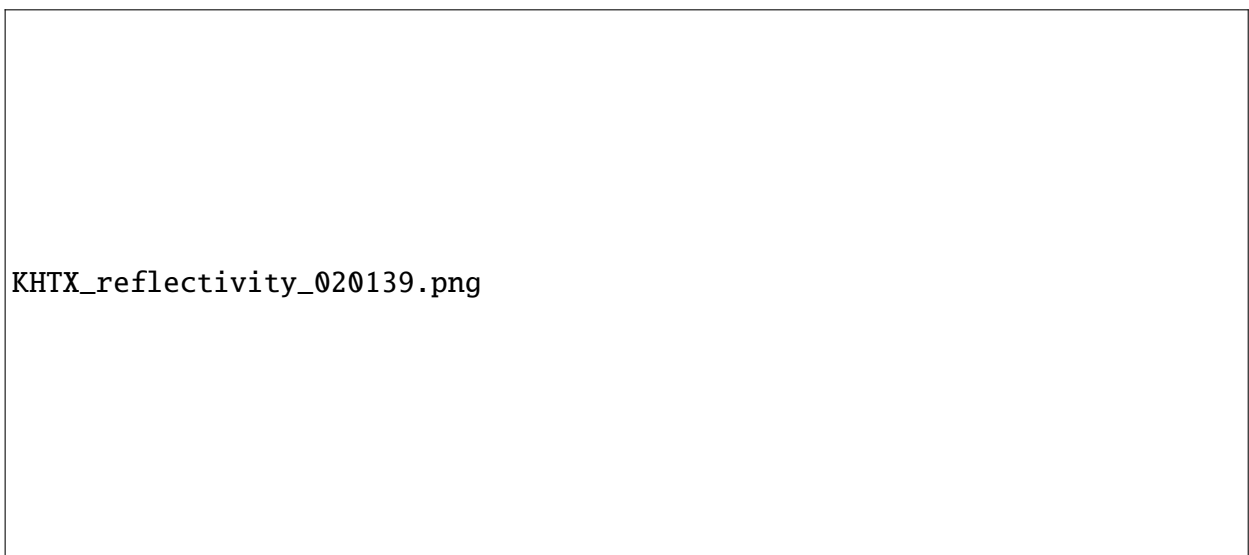


273 FIG. 4. A scatter plot of max differential reflectivity vs max flash area of each individual cell. Dots are colored
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radar_differential_reflectivity_mean-vs-max_flash_rate_ver1.png

FIG. 5. Same as Fig. 3, but for max flash rate instead of flash area.



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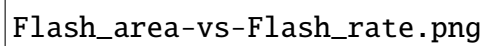
A large rectangular box containing the text "Flash_area-vs-Flash_rate.png". This box likely represents a missing figure or a placeholder for a plot showing the relationship between flash area and flash rate.

FIG. 7. Enter Caption